

Name: _____ Maths Class Teacher: _____

SYDNEY TECHNICAL HIGH SCHOOL



Extension 2 Mathematics

Trial HSC

August 2013

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using **black or blue pen**
- Board-approved calculators may be used
- **All necessary working** should be shown in question 11 -16 . Marks may be deducted for careless or badly arranged work.
- Start each question on a **new page**.
- Place your papers **in order** with the question paper on top and staple or pin them.

Total Marks - 100

Section 1 – Multiple Choice

10 marks

Attempt Questions 1 – 10

Allow 15 minutes for this section

Section 11

90 Marks

Attempt Questions 11 – 16

Allow about 2 hours 45 minutes for this section

Section 1

10 marks

Attempt Questions 1 – 10

Allow about 15 minutes for this section

Select the alternate A, B, C or D that best answers the question and indicate your choice on your multiple choice answer sheet.

1. $z_1 = 1 + 2i$ and $z_2 = 3 - i$.

The value of $z_1^2 \div \bar{z}_2$

- A. $\frac{19+7i}{10}$
B. $\frac{-5+15i}{8}$
C. $\frac{1+3i}{2}$
D. $\frac{3i-1}{2}$
-

2. By considering the graphs of $y = 3x^2 - 2x - 2$ and $y = |3x|$ the solution to $3x^2 - 2x - 2 \leq |3x|$ is,

- A. $-\frac{1}{3} \leq x \leq 2$
B. $-1 \leq x \leq \frac{3}{2}$
C. $-\frac{1}{3} \leq x \leq \frac{3}{2}$
D. $-1 \leq x \leq 2$
-

3. Consider the hyperbola with the equation $\frac{x^2}{16} - \frac{y^2}{9} = 1$.

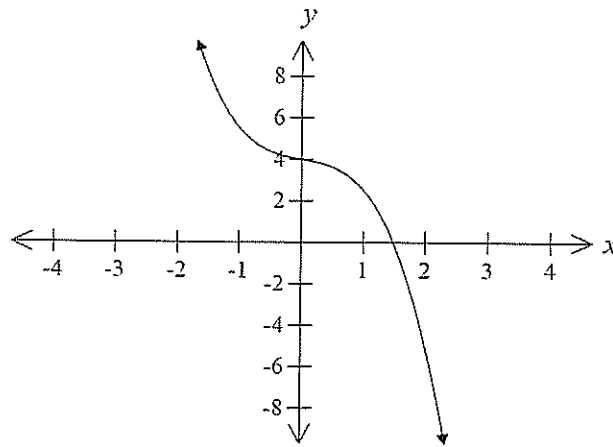
What are the coordinates of the foci of the hyperbola?

- (A) $(\pm 4, 0)$ (B) $(0, \pm 4)$
(C) $(0, \pm 5)$ (D) $(\pm 5, 0)$
-

4. The roots of $x^3 + 5x^2 + 11 = 0$ are α, β and γ , the value of $\alpha^2 + \beta^2 + \gamma^2$ is

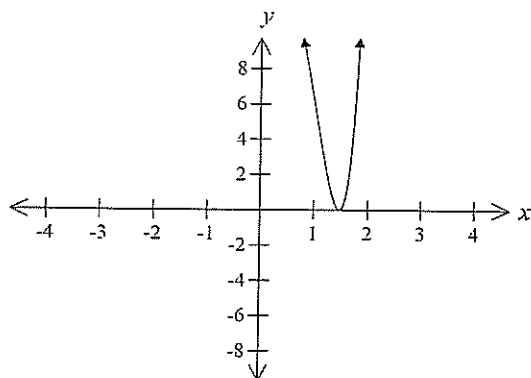
- A. -5
B. 25
C. 0
D. 3

5. The diagram below shows the graph of the function $y = f(x)$.

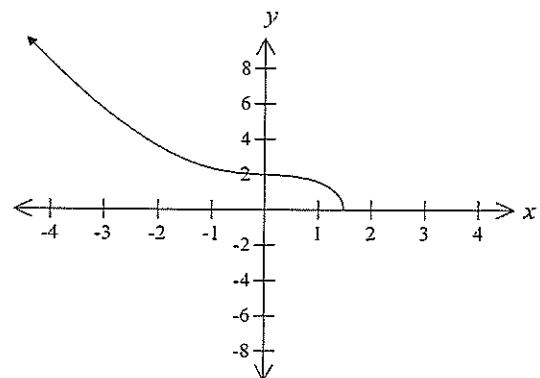


Which diagram represents the graph of $y^2 = f(x)$?

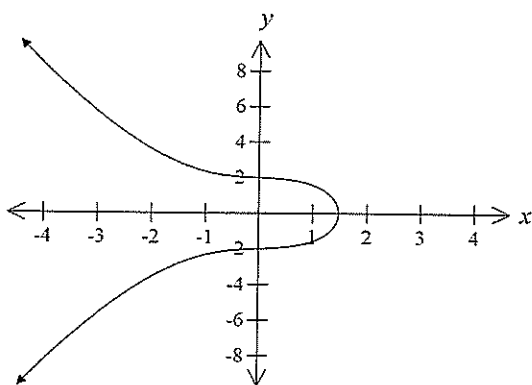
A



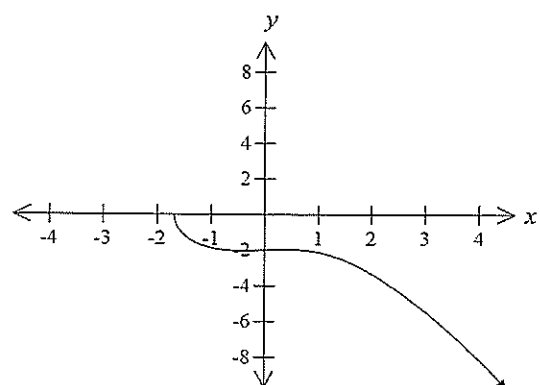
B



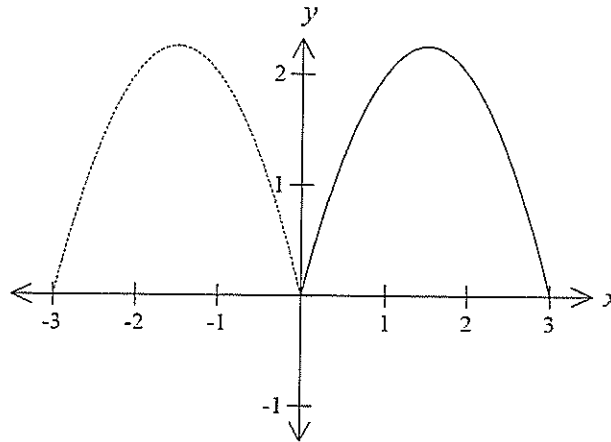
C



D



6. The area between the curve $y = 3x - x^2$, the x -axis, $x = 0$ and $x = 3$, is rotated about the y -axis to form a solid.



The volume of this solid can be found by using the integral with the method of cylindrical shells?

- A $6\pi \int_0^3 \sqrt{9 - 4y} \, dy$
- B $\pi \int_0^3 (3x^2 - x^3)^2 \, dx$
- C $2\pi \int_{-3}^3 3x^2 - x^3 \, dx$
- D $2\pi \int_0^3 3x^2 - x^3 \, dx$

7. The polynomial equation $x^3 - 3x^2 - x + 2 = 0$ has roots α , β and γ . Which of the following polynomial equations have roots $\frac{1}{\alpha}$, $\frac{1}{\beta}$ and $\frac{1}{\gamma}$?

- A $x^3 - x^2 - 3x + 1 = 0$
- B $x^3 - 2x^2 - 3x + 1 = 0$
- C $2x^3 - x^2 - 3x + 1 = 0$
- D $2x^3 - 2x^2 - 3x + 1 = 0$

8. What is the derivative of $\sin^{-1} x - \sqrt{1 - x^2}$?

- A $\frac{\sqrt{1+x}}{\sqrt{1-x}}$
- B $\frac{\sqrt{1+x}}{1-x}$
- C $\frac{1+x}{\sqrt{1-x}}$
- D $\frac{1+x}{1-x}$

9. If $p + q = 1$ and $p^2 + q^2 = 2$, the value of $p^3 + q^3$ is

- A. $1\frac{1}{2}$
 - B. $2\frac{1}{2}$
 - C. $3\frac{1}{2}$
 - D. 2
-

10. Which of the following statements is incorrect?

A $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sin^7 \theta \, d\theta = 0$

B $\int_{-1}^1 e^{-x^2} \, dx = 0$

C $\int_0^{\frac{\pi}{2}} \sin^8 \theta - \cos^8 \theta \, d\theta = 0$

D $\int_{-2}^2 \frac{x^3}{1+x^2} \, dx = 0$

End of section 1

Section II

90 Marks

Attempt Questions 11 – 16

Allow about 2 hours 45 minutes for this section

START EACH QUESTION ON A NEW PAGE

Question 11

Start a new page

a. Find,

i. $\int \frac{4x-16}{x^2-8x+20} dx$ 2

ii. $\int \frac{1}{x^2-8x+20} dx$ 2

iii. $\int \frac{x}{\sqrt{x-1}} dx$ 3

b. Evaluate,

i. $\int_0^{0.5} \sin^{-1} x dx$ 2

ii. $\int_0^{\frac{\pi}{4}} x \sec^2 x dx$ 2

c. i. Find the value of **a** and **b** such that

$$\frac{x}{(x+1)(x+2)} = \frac{a}{x+1} + \frac{b}{x+2} \quad 2$$

ii. Hence, evaluate $\int_0^1 \frac{x}{(x+1)(x+2)} dx$ 2

Question 12

Start a new page

- a.
- i. Find the two square roots of $2i$ 3
 - ii. Solve $x^2 + 2x + \left(1 - \frac{i}{2}\right) = 0$ 2
- b. Find α and β given that $z^3 + 3z + 2i = (z - \alpha)^2(z - \beta)$ 2
- c.
- i. On a Argand diagram, sketch the locus of the point P representing the complex number z which moves so that $|z - 2| = 1$ 2
 - ii. Find the range of possible values of $|z|$ and $\arg(z)$ 2
 - iii. The points P_1 and P_2 such that OP_1 and OP_2 are tangents to the locus, (O is the origin) represent the complex numbers z_1 and z_2 respectively. Express z_1 and z_2 in modulus – argument form 2
 - iv. Evaluate $z_1^{20} + z_2^{20}$, give your answer in its simplest form. 2
-

Question 13

Start a new page

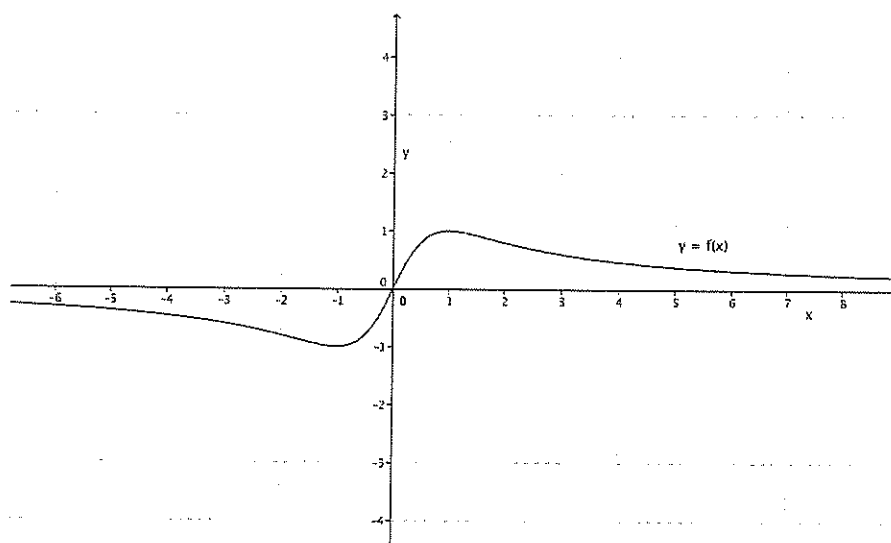
a. Evaluate $\int_0^{\frac{\pi}{2}} \frac{d\theta}{2 + \cos \theta}$ by using the substitution $t = \tan \frac{\theta}{2}$ 3

b.

i. If $I_n = \int_0^1 x^n e^{-x} dx$, where n is a positive integer, show that $I_n = nI_{n-1} - \frac{1}{e}$ 2

ii. Hence, evaluate $\int_0^1 x^3 e^{-x} dx$ 2

c. The diagram below is of the function $f(x) = \frac{2x}{x^2+1}$



Sketch the following on separate number planes, without the use of calculus 8

i. $y = f(|x|)$

ii. $|y| = f(x)$

iii. $y \times f(x) = 1$

iv. $y = e^{f(x)}$

Question 14

Start a new page

- a. $P(x) = x^4 - x^3 - 2x^2 + 6x - 4$. Given that $1 + i$ is a zero of $P(x)$, find all the zeros of $P(x)$

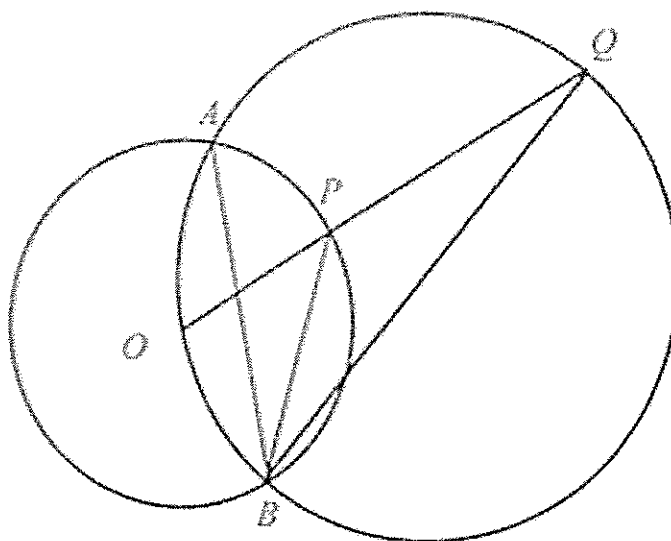
3

- b. Two sides of a triangle are in the ratio 3 : 1 and the angles opposite these sides differ by $\frac{\pi}{6}$. Show that the smaller of the two angles is,

4

$$\tan^{-1} \frac{1}{6 - \sqrt{3}}$$

c.



In the diagram above, the centre O of the small circle APB lies on the circumference of the larger circle AQB . The points O, P and Q are collinear.

- i. Let angle $OAB = x$, show that angle $OQB = x$ 1
- ii. Let angle $ABP = y$, find an expression for angle OPB 2
- iii. Prove that BP bisects $\angle ABQ$ 2

- d. Show that the polynomial $P(x) = x^n - x^{n-1} - 1$, where $n > 1$ cannot have a repeated root.

3

Question 15

Start a new page

a.

- i. Show that the ellipse $4x^2 + 9y^2 = 36$ and the hyperbola $4x^2 - y^2 = 4$ intersect at right angles, in the first quadrant.

3

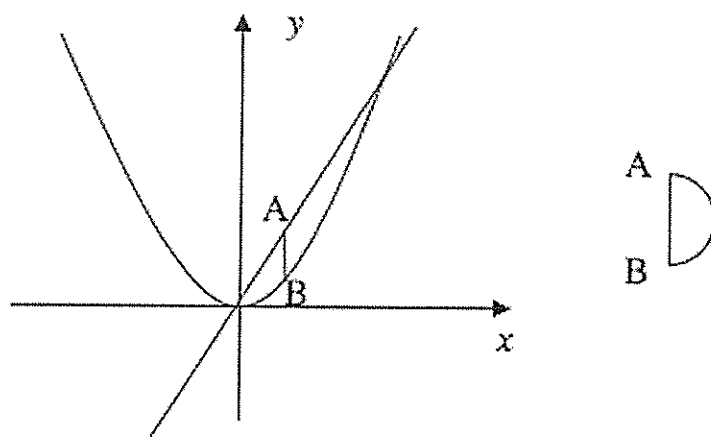
- ii. Find the equation of the circle through the points of intersection of the two conics.

1

- b. The base of a solid is the region enclosed by $y = 2x$ and $y = x^2$.

5

Cross sections taken perpendicular to the x - axis are semi - circles with the diameter in the base of the solid (as indicated the diameter AB of the semicircle is perpendicular to the x axis; the semicircle is perpendicular to the xy plane)



Find the volume of the solid

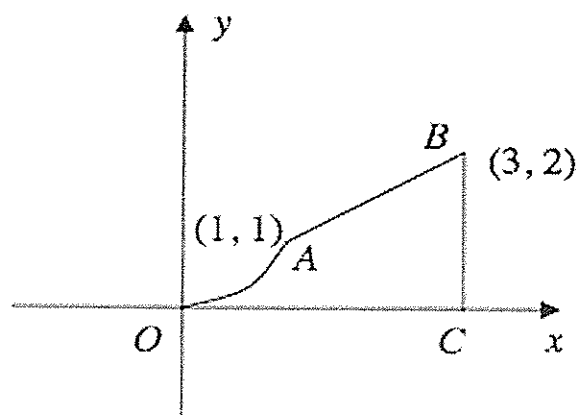
- c. OA is an arc of the parabola $y = x^2$. The region OABC is rotated about the y - axis forming a bowl.

- i. By using cylindrical shells determine the volume of the solid formed

5

- ii. Hence, find the holding capacity of the bowl.

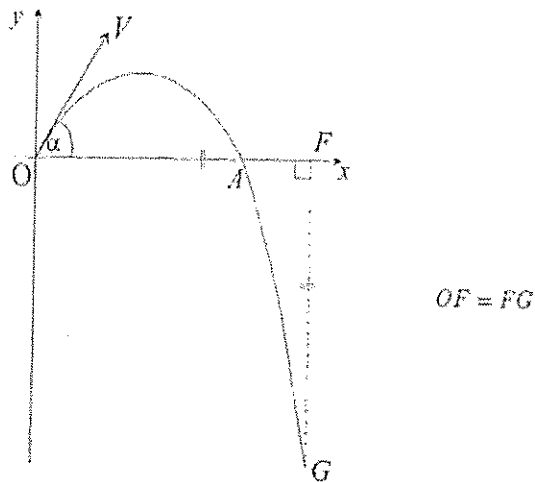
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Question 16

Start a new page

a.



In the diagram above, a projectile is fired from a point O at the top of a vertical cliff.

Its initial speed is V m/s and its angle of elevation is α . Let the acceleration due to gravity be g m/s².

- i. By using the equation of motion $\ddot{x} = 0$ and $\ddot{y} = -g$, derive the expressions for the horizontal and vertical displacements after t seconds. 2
- ii. Let G be the point on the projectile's path where the distance below the origin equals the distance to the right of the origin. That is, $OF = FG$ on the diagram above.

α . Prove that the time taken for the projectile to reach G is

$$\frac{2V(\sin \alpha + \cos \alpha)}{g} \text{ seconds.} \quad 2$$

β . Show that $OF = \frac{v^2}{g}(\sin 2\alpha + \cos 2\alpha + 1)$ metres. 2

γ . Let A be the point on the projectile's path where it is level with the point of projection. 4

If $OF = \frac{4}{3}OA$, find α , to the nearest degree.

You may assume that the distance OA is given by $OA = \frac{v^2 \sin 2\alpha}{g}$ metres.

b. Define $f^{(n)}(x)$ to be $f(f \dots (f(x)) \dots)$ where f is repeated n times

5

That is,

$$f^{(1)}(x) = f(x)$$

$$f^{(2)}(x) = f(f(x)).$$

$$f^{(3)}(x) = f(f(f(x))) \text{ etc}$$

$$\text{Let } f(x) = \frac{x}{\sqrt{1+x^2}}$$

Prove by mathematical induction that $f^{(n)}(x) = \frac{x}{\sqrt{1+nx^2}}$

End of examination

Section I - Multiple Choice (1 mark each)

- | | | | | |
|------|------|------|------|-------|
| 1. D | 2. D | 3. D | 4. B | 5. C |
| 6. D | 7. C | 8. A | 9. B | 10. B |

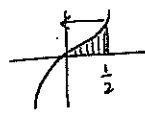
Section II

Question 11

a. i. $\int \frac{4x-16}{x^2-8x+20} dx = 2 \ln(x^2-8x+20) + C$

ii. $\int \frac{1}{x^2-8x+20} dx = \int \frac{1}{(x-4)^2+4} dx$
 $= \frac{1}{2} \tan^{-1}\left(\frac{x-4}{2}\right) + C$

iii. $\int \frac{x}{\sqrt{x-1}} dx$ let $x-1=u^2$ $u=\sqrt{x-1}$
 $x=u^2+1$
 $dx=2u \cdot du$
 $\int \frac{u^2+1}{\sqrt{u^2}} \cdot 2u du$
 $= 2 \int u^2+1 du$
 $= 2 \left[\frac{u^3}{3} + u \right]$
 $= \frac{2}{3}(x-1)\sqrt{x-1} + 2\sqrt{x-1} + C$

b. i. $\int_0^{0.5} \sin^{-1} x dx$ 
 $= \frac{\pi}{6} \times \frac{1}{2} - \int_0^{\pi/6} \sin y dy$
 $= \frac{\pi}{12} + [\cos y]_0^{\pi/6}$
 $= \frac{\pi}{12} + \frac{\sqrt{3}}{2} - 1$
 $= \frac{\pi + 6\sqrt{3} - 12}{12}$

ii. $\int_0^{\pi/4} x \sec^2 x dx$
 $= x \tan x - \int 1 \tan x dx$
 $= \left[x \tan x + \ln(\cos x) \right]_0^{\pi/4}$
 $= \left[\frac{\pi}{4} \times 1 + \ln\left(\frac{1}{\sqrt{2}}\right) - (0 + \ln 1) \right]$
 $= \frac{\pi}{4} - \frac{1}{2} \ln 2$

c. i. $\frac{x}{(x+1)(x+2)} = \frac{a}{x+1} + \frac{b}{x+2}$
 $x = a(x+2) + b(x+1)$
 let $x = -1$ $-1 = a(1)$ $\therefore a = -1$
 let $x = -2$ $-2 = b(-1)$ $\therefore b = 2$

ii. $\int_0^1 \frac{-1}{x+1} + \frac{2}{x+2} dx$
 $= -\ln(x+1) + 2 \ln(x+2) \Big|_0^1$
 $= 2 \ln(3) - \ln(2) - [2 \ln(2) - 0]$
 $= 2 \ln 3 - 3 \ln 2$
 $= \ln 9 - \ln 8$
 $= \ln(9/8)$

a. 1. $\sqrt{2i} = x + iy$

$$2i = (x + iy)^2$$

$$= x^2 + 2xyi - y^2$$

equating reals: $x^2 - y^2 = 0$

Im: $2xy = 2$
 $xy = 1$

By inspection $x=1$ or $x=-1$
 $y=1$ or $y=-1$

$\therefore \sqrt{2i} = 1+i$ or $-1-i$

11. $x^2 + 2x + (1 - i/2) = 0$

$$x = \frac{-2 \pm \sqrt{4 - 4 \times 1 \times (1 - i/2)}}{2}$$

$$= \frac{-2 \pm \sqrt{2i}}{2}$$

) $= \frac{-2 \pm (x+i)}{2}$ or $\frac{-2 \pm (-x-i)}{2}$

$x = \frac{-1+i}{2}$ or $\frac{-3-i}{2}$

b. $z^3 + 3z + 2i = (z - \alpha)^2 (z - \beta)$

sum: $2\alpha + \beta = \frac{-b}{a} = 0$ — ①

double: $\alpha^2 + 2\alpha\beta = \frac{c}{a} = 3$ — ②

triple: $\alpha^2\beta = -2i$ — ③

Solving ① and ②

$$2\alpha\beta + \alpha^2 = 3$$

$$2\alpha(-2\alpha) + \alpha^2 = 3$$

$$-3\alpha^2 = 3$$

$$\alpha^2 = -1$$

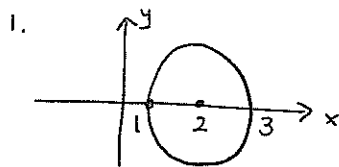
$$\alpha = \pm i \therefore \beta = \mp 2i$$

$\alpha = -i$ $\beta = 2i$ works when tested

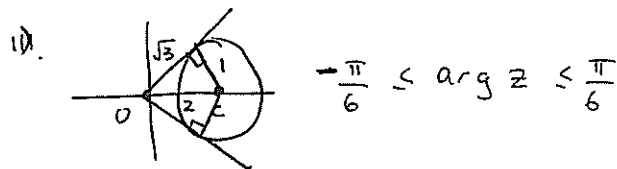
($\alpha = i$ and $\beta = -2i$)
doesn't

$\therefore \alpha = -i, \beta = 2i$

c. $|z - 2| = 1$



1). $1 \leq |z| \leq 3$ (using values on x-axis)



(iii) $P_1 \rightarrow z_1 = r \operatorname{cis} \theta_1$

$= \sqrt{3} \operatorname{cis} \frac{\pi}{6}$ ← answer here

$= \left(\frac{3}{2} + \frac{\sqrt{3}i}{2} \right)$ ← can accept if u want

$P_2 \rightarrow z_2 = r \operatorname{cis} \theta_2$

$= \sqrt{3} \operatorname{cis} \left(-\frac{\pi}{6} \right)$ ← answer

$= \left(\frac{3}{2} - \frac{\sqrt{3}i}{2} \right)$

iv) $z_1^{20} + z_2^{20} = \left(\sqrt{3} \operatorname{cis} \frac{\pi}{6} \right)^{20} + \left(\sqrt{3} \operatorname{cis} -\frac{\pi}{6} \right)^{20}$

$= \sqrt{3}^{20} \operatorname{cis} \frac{20\pi}{6} + \sqrt{3}^{20} \operatorname{cis} \left(-\frac{20\pi}{6} \right)$

$= -3^{10}$

Question 13

$$a. \int_0^{\pi/2} \frac{d\theta}{2 + \cos\theta} \quad \text{let } t = \tan\frac{\theta}{2}$$

$$= \int_0^1 \frac{1}{2 + \frac{1-t^2}{1+t^2}} \cdot \frac{2dt}{t^2+1} \quad \begin{matrix} d\theta = \frac{2dt}{t^2+1} \\ \theta = \pi/2 \quad t=1 \\ \theta = 0 \quad t=0 \end{matrix}$$

$$= \int_0^1 \frac{2dt}{2(1+t^2) + 1 - t^2}$$

$$= \int_0^1 \frac{2dt}{3 + t^2}$$

$$= \frac{2}{\sqrt{3}} \left[\tan^{-1}\left(\frac{t}{\sqrt{3}}\right) \right]_0^1$$

$$= \frac{2}{\sqrt{3}} \left[\frac{\pi}{6} - 0 \right]$$

$$= \frac{\pi}{3\sqrt{3}}$$

$$b. I_n = \int_0^1 x^n e^{-x} dx$$

$$I_n = \left[\frac{x^n \cdot e^{-x}}{-1} \right]_0^1 - \int_0^1 \frac{n x^{n-1} \cdot e^{-x}}{-1} dx$$

$$= \left[-1^n e^{-1} - 0 \right] + n \int_0^1 x^{n-1} e^{-x} dx$$

$$= -\frac{1}{e} + n I_{n-1}$$

$$= n I_{n-1} - \frac{1}{e} \quad \text{as req.}$$

$$ii. I_3 = 3 I_2 - \frac{1}{e}$$

$$= 3 \left[2 I_1 - \frac{1}{e} \right] - \frac{1}{e}$$

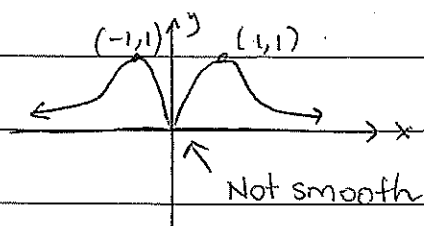
$$= 3 \left[2 \left(I_0 - \frac{1}{e} \right) - \frac{1}{e} \right] - \frac{1}{e}$$

$$I_0 = \int_0^1 e^{-x} dx = 1 - \frac{1}{e}$$

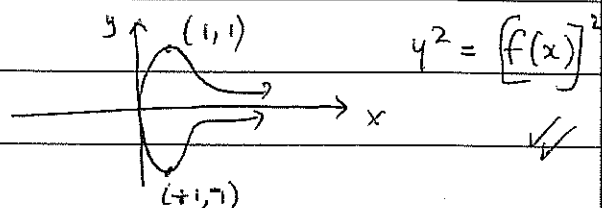
$$\therefore I_3 = 3 \left[2 - \frac{3}{e} \right] - \frac{1}{e}$$

$$= 6 - \frac{16}{e}$$

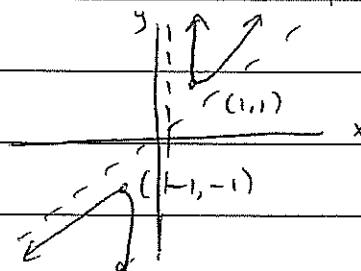
$$c. i. y = f|x|$$



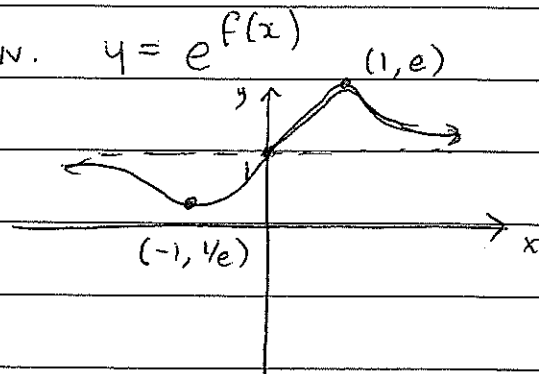
$$ii. |y| = f(x) \quad \therefore f(x) \geq 0$$



$$iii. y \times f(x) = 1$$



$$iv. y = e^{f(x)}$$



Question 14.

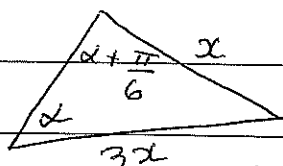
a. $P(x) = x^4 - x^3 - 2x^2 + 6x - 4$

if $1+i$ is a factor so is $1-i$ (conjugate pairs) $\therefore x^2 - 2x + 2$ is a factor

$$x^2 + x - 2$$

$$(x^2 - 2x + 2)(x^4 - x^3 - 2x^2 + 6x - 4)$$

$$P(x) = (x+2)(x-1)(x^2 - 2x + 2)$$

zeros are $-2, 1, 1 \pm i$ b. let smaller angle be α 

Sine rule

$$\frac{\sin(\alpha + \pi/6)}{3x} = \frac{\sin \alpha}{x}$$

$$\sin(\alpha + \pi/6) = 3 \sin \alpha$$

$$\sin \alpha \frac{\sqrt{3}}{2} + \cos \alpha \cdot \frac{1}{2} = 3 \sin \alpha$$

$$\cos \alpha = 2 \sin \alpha (3 - \frac{\sqrt{3}}{2})$$

$$\cot \alpha = 6 - \sqrt{3}$$

$$\therefore \tan \alpha = \frac{1}{6 - \sqrt{3}}$$

$$6 - \sqrt{3}$$

$$\therefore \alpha = \tan^{-1} \left(\frac{1}{6 - \sqrt{3}} \right)$$

c. let $\angle OAB = x$

$$\angle OQB = x$$
 (angle at circumference on arc AB circle AQB)

 $OA = OB$ radii $\therefore \triangle OAB$ isisosceles with $\angle OAB = \angle OBA$

$$\therefore \angle OBA = x$$

ii. $\angle ABP = y$

$$\therefore \angle OBP = x + y$$

 $OP = OB$ radii circle centre O

$$\angle OPB = \angle OBP$$

$$= x + y$$

iii. $\angle OPB = \angle PQB + \angle PBO$

$$\therefore \angle PBQ = y$$

and $\angle PBQ = \angle ABP (= y)$

 $\therefore PB$ bisects $\angle ABQ$.

d. $P(x) = x^n - x^{n-1} - 1, n > 1$

repeated root when $P(x) = P'(x) = 0$

$$P'(x) = nx^{n-1} - (n-1)x^{n-2} \text{ if } P'(x) = 0$$

$$nx^{n-1} - (n-1)x^{n-2} = 0$$

$$x^{n-2} [xn - n + 1] = 0$$

$$\downarrow$$

 $x = 0$

$$xn = n - 1$$

but $P(x) \neq 0$

$$x = \frac{n-1}{n}$$

Now $P\left(\frac{n-1}{n}\right) = \left(\frac{n-1}{n}\right)^n - \left(\frac{n-1}{n}\right)^{n-1} - 1$

$$0 = \frac{(n-1)^{n-1} [n-1-n] - 1}{n^n}$$

$$1 = \frac{-1(n-1)^{n-1}}{n^n} \text{ but } n > 1$$

$$\therefore \frac{-1(n-1)^{n-1}}{n^n} < 0 \therefore$$

$$x \neq \frac{n-1}{n}$$

 $\therefore$ can't have a repeated root.

a. $4x^2 + 9y^2 = 36$ and $4x^2 - y^2 = 4$
 ① ②

$$10y^2 = 32$$

$$y^2 = 3.2 \quad y = \pm \sqrt{3.2} = \pm 4\sqrt{0.2}$$

$$4x^2 - y^2 = 4 \Rightarrow 4x^2 - 3.2 = 4$$

$$4x^2 = 7.2$$

$$x^2 = 1.8$$

$$x = \pm 3\sqrt{0.2}$$

For $4x^2 + 9y^2 = 36$

$$8x + 18y \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = \frac{-4x}{9y} \quad \text{at } (\pm 3\sqrt{0.2}, \pm 4\sqrt{0.2})$$

take $(3\sqrt{0.2}, 4\sqrt{0.2})$
 1st quad

$$M_T = -\frac{1}{3}$$

For $4x^2 - y^2 = 4$

$$8x - 2y \frac{dy}{dx} = 0$$

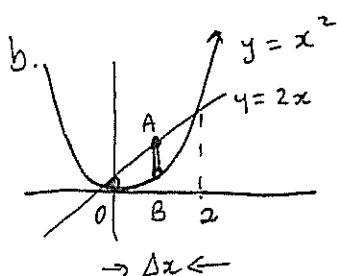
$$\frac{dy}{dx} = \frac{4x}{y} \quad \text{at } (3\sqrt{0.2}, 4\sqrt{0.2})$$

$$M_T = 3$$

∵ as gradients of tangents are negative reciprocals tangents are $\perp$.

11 distance from $(0,0)$ to $(3\sqrt{0.2}, 4\sqrt{0.2})$
 $= 5$ (this is the radius of circle as dist $\perp$)

∵ Locus $x^2 + y^2 = 5$



diameter $2x - x^2$
 radius $= x - \frac{1}{2}x^2$

Area cross section

$$A = \frac{1}{2} \pi r^2$$

$$= \frac{1}{2} \pi \left(x - \frac{1}{2}x^2 \right)^2$$

$$\Delta V = \frac{1}{2} \pi \left(x - \frac{1}{2}x^2 \right)^2 \Delta x$$

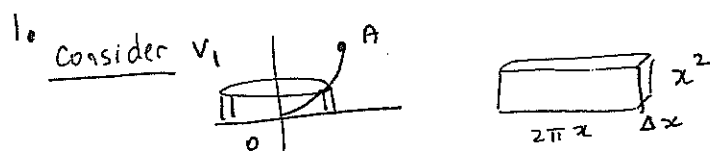
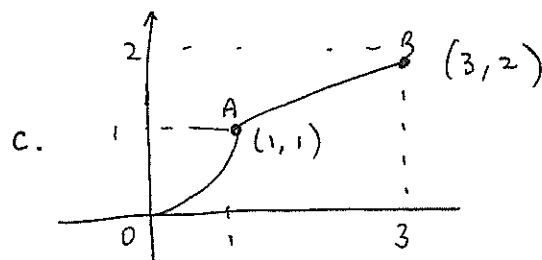
Total Volume $= \lim_{\Delta x \rightarrow 0} \sum_{x=0}^2 \frac{\pi}{2} \left(x - \frac{x^2}{2} \right)^2 \Delta x$

$$V = \frac{\pi}{2} \int_0^2 \left(x^2 - x^3 + \frac{x^4}{4} \right) dx$$

$$= \frac{\pi}{2} \left[\frac{x^3}{3} - \frac{x^4}{4} + \frac{x^5}{20} \right]_0^2$$

$$= \frac{\pi}{2} \left[\frac{8}{3} - 4 + \frac{32}{20} - (0) \right]$$

$$= \frac{2\pi}{15} u^3$$



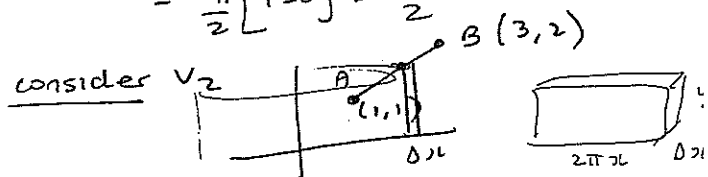
$$\Delta V = 2\pi x^3 \Delta x$$

$$\therefore V_1 = \lim_{\Delta x \rightarrow 0} \sum_0^1 2\pi x^3 \Delta x$$

$$= 2\pi \int_0^1 x^3 dx$$

$$= \frac{2\pi}{4} [x^4]_0^1$$

$$= \frac{\pi}{2} [1 - 0] = \frac{\pi}{2} u^3$$



consider V_2 equation Line AB $\Rightarrow y = \frac{x+1}{2}$

$$\Delta V = 2\pi x \left[\frac{x+1}{2} \right] \Delta x$$

$$V_2 = \lim_{\Delta x \rightarrow 0} \sum_1^3 \pi x \left(\frac{x+1}{2} \right) \Delta x$$

$$= \pi \int_1^3 \left(x^2 + x \right) dx$$

$$= \pi \left[\frac{x^3}{3} + \frac{x^2}{2} \right]_1^3$$

$$= \pi \left[9 + \frac{9}{2} - \left(\frac{1}{3} + \frac{1}{2} \right) \right] = \frac{38\pi}{3}$$

∵ Volume bowl $= \frac{\pi}{2} + \frac{38\pi}{3} = \frac{79\pi}{6}$

11. Capacity $= \pi R^2 H - \frac{79\pi}{6}$
 $= 18\pi - \frac{79\pi}{6} = \frac{29\pi}{6} u^3$

2. H
1. $\ddot{x} = 0$

$\dot{x} = C_1$

$t=0 \dot{x} = v \cos \alpha$

$\dot{x} = v \cos \alpha$

$x = v \cos \alpha t + C_2$

$t=0 x=0 \therefore C_2=0$

$x = v \cos \alpha t$

$\ddot{y} = -g$

$\dot{y} = -gt + C_3$ $t=0$

$\dot{y} = v \sin \alpha$

$\dot{y} = -gt + v \sin \alpha$

$y = -g \frac{t^2}{2} + v \sin \alpha t + C_4$

$t=0 y=0$

$\therefore C_4=0$

$y = v \sin \alpha t - \frac{gt^2}{2}$

11. OA = FQ hence

1) $v \sin \alpha t - \frac{gt^2}{2} = -v \cos \alpha t$

$\frac{gt^2}{2} = v \sin \alpha t + v \cos \alpha t$

$\div t$ as $t \neq 0$

$\frac{gt}{2} = v \sin \alpha + v \cos \alpha$

$t = \frac{2v}{g} (\sin \alpha + \cos \alpha)$ sec.

2) DF = $v \cos \alpha t$

$= v \cos \alpha \left[\frac{2v}{g} (\sin \alpha + \cos \alpha) \right]$

$= \frac{v^2}{g} (2 \sin \alpha \cos \alpha + 2 \cos^2 \alpha)$

$= \frac{v^2}{g} (\sin 2\alpha + \cos 2\alpha + 1)$ m

3) Several solⁿ possible:

DF = $\frac{4}{3}$ OA

$\frac{v^2}{g} (\sin 2\alpha + \cos 2\alpha + 1) = \frac{4}{3} \frac{v^2}{g} \sin 2\alpha$

$\sin 2\alpha + \cos 2\alpha + 1 = \frac{4}{3} \sin 2\alpha$

$\cos 2\alpha + 1 = \frac{1}{3} \sin 2\alpha$

$\sin 2\alpha = 3 \cos 2\alpha + 3$

ie $\sin 2\alpha - 3 \cos 2\alpha = 3$

let $\sin 2\alpha - 3 \cos 2\alpha = R \sin (2\alpha - \theta)$

$= \sqrt{10} \sin (2\alpha - \tan^{-1}(3))$

$= \sqrt{10} \sin (2\alpha - 71^\circ 34')$

$\sqrt{10} \sin (2\alpha - 71^\circ 34') = 3$

$2\alpha - 71^\circ 34' = \sin^{-1}(\frac{3}{\sqrt{10}})$

$\alpha = 71^\circ 34' \div 2$ (near est degree)

b. $f^n(x) = \frac{x}{\sqrt{1+n x^2}}$ $f(x) = \frac{x}{\sqrt{1+x^2}}$

test $n=1$

$f'(x) = \frac{x}{\sqrt{1+x^2}}$

$= \frac{x}{\sqrt{1+x^2}}$

$= f(x) \therefore$ true for $n=1$

Assume true for $n=k$

ie $f^k(x) = f(f(f \dots f(x))) = \frac{x}{\sqrt{1+k x^2}}$

Prove true for $n=k+1$

ie $f^{k+1}(x) = f(f(f(f \dots f(x))))$

$\xleftarrow{k+1}$

$= f[f^k(x)]$ times

$= f\left[\frac{x}{\sqrt{1+k x^2}}\right]$

$= \frac{x}{\sqrt{1+k x^2}}$

$= \frac{x}{\sqrt{1 + \left(\frac{x}{\sqrt{1+k x^2}}\right)^2}}$

$=$

$= \frac{x}{\sqrt{1+k x^2}}$

$= \frac{x}{\sqrt{1 + \frac{x^2}{1+k x^2}}}$

$= \frac{x}{\sqrt{1+k x^2} \sqrt{1 + \frac{x^2}{1+k x^2}}}$

$= \frac{x}{\sqrt{1+k x^2 + \frac{x^2(1+k x^2)}{1+k x^2}}}$

$= \frac{x}{\sqrt{1+k x^2 + x^2}}$

$= \frac{x}{\sqrt{1+(k+1)x^2}}$

$\therefore$ true $n=k+1$ if true $n=k$.

ie. As true for $n=1$, also true for $n=2, 3, 4$ etc hence by M.I true all positive integer n .